ARE HUMAN-CONSTRAINED LANGUAGE MODELS MORE HUMAN-LIKE? EVALUATING ALGORITHMIC CONVERGENCE ACROSS NEURAL AND BEHAVIORAL MEASURES

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ABSTRACT

Do Large Language Models (LLMs) and biological brains converge onto similar computational mechanisms during language processing? Applying Marr's tri-level framework reveals substantial mismatches between humans and LLMs at both the computational level (broad communicative intent versus next-token prediction) and the implementational level (biological neural circuits versus artificial Transformer/SSM architectures). Whether these systems nevertheless converge at the algorithmic level – the representations and step-by-step processing dynamics – remains an open question. To test for algorithmic alignment, cognitive scientists have increasingly developed "human-like" models constrained by human cognitive limitations (e.g., working memory bottlenecks) or developmentally plausible training exposure (e.g., the BabyLM challenge). Here, we evaluate algorithmic convergence by establishing a multi-modal human benchmark, evaluating how accurately model-derived metrics (surprisal and internal contextual embeddings) predict human behavioral responses (reading times and eye-tracking fixations) and brain activity (fMRI, MEG, EEG event-related potentials, and intracranial ECoG recordings). Counter to the hypothesis that human-like inductive biases guarantee greater cognitive alignment, our results demonstrate that current human-like models do not outperform standard, unconstrained LLMs in predicting human neural and behavioral metrics at the algorithmic level. Instead, the strong alignment observed in standard LLMs despite implementational differences lends empirical support to the Platonic Representation Hypothesis, suggesting that task-scale optimization naturally drives disparate information-processing systems toward shared representations. We make available our multi-modal evaluation framework as the LLM-Human Convergence Benchmark, providing a systematic testing ground to guide the future design of genuinely human-aligned cognitive architectures.

1 INTRODUCTION

Do humans and Large Language Models (LLMs) process language in a similar way? David Marr's tri-level framework provides a principled lens to address this question by decomposing information-processing systems into three distinct levels (Figure 1A): the computational level (what is the goal of the computation?), the algorithmic level (what representations and step-by-step procedures are used?), and the implementational level (how are these steps physically realized?) (Marr, 1982). To evaluate whether LLMs serve as meaningful models of human cognition, we can analyze both systems across these three levels. At the computational level, LLMs are primarily optimized for next-word prediction over massive text corpora. But is next-token prediction truly what the human brain does? While predictive processing plays a central role in human language (Hale, 2001; Levy, 2008; Smith & Levy, 2013; Shain et al., 2024), next-token prediction is at best a rough proxy; biological language processing is driven by broader communicative intent, pragmatic inferencing, multi-step predictive horizons, and complex social or goal-directed rewards (Bender & Koller, 2020; Andreas, 2022). At the implementational level, clear physical and structural distinctions seem to exist. Although artificial neural networks drew early inspiration from neuroscience, modern LLMs

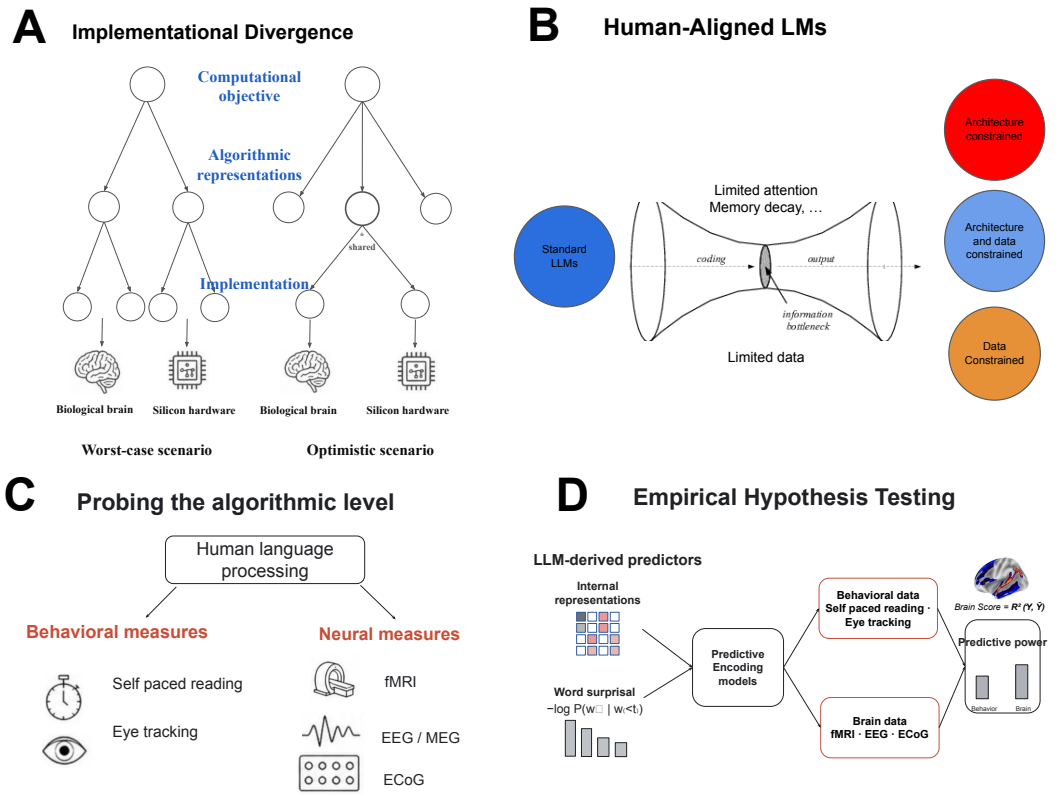


Figure 1: Overview of the study. (A) Marr's tri-level framework applied to humans and LLMs: even where the two systems share the same algorithmic-level representations (optimistic scenario), they may still diverge at the computational and implementational levels. (B) Human-aligned language models introduce either architectural constraints (limited attention, memory decay) or data constraints (developmentally limited training exposure) relative to standard LLMs. (C) Human language processing is probed through behavioral measures (self-paced reading, eye-tracking) and neural measures (fMRI, EEG/MEG, ECoG) at the algorithmic level. (D) LLM-derived predictors – internal representations and word surprisal – are fed through predictive encoding models to obtain BehavScore and BrainScore, which quantify predictive power over behavioral and neural data respectively.

have diverged significantly from their biological counterparts. Standard LLMs rely on global self-attention mechanisms across vast context windows (Vaswani et al., 2017), uniform layerwise Transformer blocks that lack the dedicated modularity of the human cortical language network, non-local backpropagation rather than biologically plausible local learning rules (e.g., STDP), and training exposures spanning trillions of tokens – orders of magnitude beyond a human lifetime (Gilkerson et al., 2017; Warstadt et al., 2023a).

Given these differences at both the computational level (Level 1) and implementational level (Level 3), do LLMs and humans converge onto similar representations and processing steps at the algorithmic level (Level 2)? One can map out a spectrum of potential outcomes regarding their algorithmic alignment (Figure 1A). At the pessimistic extreme, these mismatches drive LLMs and human brains to adopt fundamentally different algorithmic steps to solve linguistic tasks. At the optimistic extreme – aligned with the Platonic Representation Hypothesis (Huh et al., 2024) – shared environmental pressures and task demands force biological and artificial systems to converge onto the same underlying representations and algorithms despite their structural differences. Whether LLMs and human brains actually converge at the algorithmic level remains an open empirical question.

Unlike engineering goals focused purely on task performance, cognitive scientists and neuroscientists aim to build models that emulate and explain human-like cognition. To guarantee that artificial

models converge on mechanisms closer to human language processing at the algorithmic level, researchers have begun incorporating explicit biological and developmental constraints. On the architectural side, researchers introduce human cognitive limitations, such as working memory capacity bottlenecks (Lewis & Vasishth, 2005), recurrent processing dynamics (Michaelov et al., 2024a), or localized attention structures (de Varda & Marelli, 2024; Kuribayashi et al., 2022). On the data side, initiatives such as the BabyLM Challenge restrict training corpora to developmentally realistic, child-scale linguistic input (10M to 100M words) rather than web-scale text (Figure 1B; Warstadt et al., 2023a;b). However, whether these human-like constraints actually succeed in driving algorithmic similarity remains an open question. It is entirely unclear if introducing such inductive biases genuinely aligns model mechanisms with human processing, or if standard, unconstrained LLMs already achieve comparable algorithmic convergence simply through task-scale optimization.

How can we empirically determine which algorithms are used by humans and LLMs during language processing? In humans, algorithmic steps are empirically probed through behavioral and neural measures (Figure 1B). Specifically, in psycholinguistics, behavioral measures typically include error rates, error types, self-paced reading times, and eye-tracking fixations (Futrell et al., 2021; Frank et al., 2013; Luke & Christianson, 2018; Siegelman et al., 2022). These offer insights into processing difficulty and moment-by-moment cognitive load. In neuroscience, algorithmic dynamics are captured across complementary spatio-temporal scales: functional Magnetic Resonance Imaging (fMRI) provides whole-brain spatial coverage of, e.g., representational similarity (Schrimpf et al., 2021); Electroencephalography (EEG) tracks precise temporal dynamics, such as the N400 and P600 event-related potentials (Michaelov et al., 2024b; Federmeier et al., 2007); and intracranial electrophysiology (e.g., ECoG) provides joint spatial precision and millisecond-level signal resolution directly from cortical circuits (Schrimpf et al., 2021). Combined, these multi-modal measurements provide the means to evaluate the neural representations and computations underlying language processing. In artificial neural networks, internal mechanisms can be analyzed directly via model interpretability methods; however, given the immense complexity of these black-box representations, interpreting them in isolation remains challenging. Here, rather than relying on internal model interpretability, we test for algorithmic convergence by examining whether LLM-derived metrics, such as token surprisal and internal contextual embeddings, can accurately predict behavioral and neural measurements recorded from humans. Evaluating this predictive power across different model classes allows us to answer the central question of this study: whether humans and LLMs converge on similar algorithmic solutions, and in particular, whether human-constrained models achieve significantly greater convergence than standard LLMs.

We systematically test and contrast four groups of models in their capacity to predict human behavioral and neural responses (Figure 1C), crossing the architectural and data constraints above into a 2×2 design: (1) memory-constrained architectures trained at web scale, (2) the same memory-constrained architectures trained on BabyLM-scale data, (3) standard architectures trained on BabyLM-scale data, and (4) standard, off-the-shelf LLMs with neither constraint. By evaluating these four groups against our multi-modal human benchmark, we investigate whether introducing human-like constraints – in network architecture, training data size, or both – is essential for achieving genuine algorithmic alignment between artificial neural networks and the human brain, and whether the two kinds of constraint compound or substitute for one another.

Across seventeen behavioral and neural datasets and over a hundred evaluated models, we find no evidence that either constraint helps: on every dataset modality we test, the Pareto front trading off neural and behavioral predictive power is occupied exclusively by standard, unconstrained LLMs, and this holds whether we look at all words or specifically at the syntactically hardest ones. Restricting a model's attention to nearby context lowers its alignment with human data outright, regardless of whether it is trained at web scale or human scale; restricting training data alone, while it leaves a standard architecture intact, still does not produce a model that outperforms its web-scale counterpart on neural predictivity, though it comes far closer on behavioral predictivity than the architecturally constrained models do. Scale itself keeps paying off within the unconstrained group, across nearly every modality we test, in a way it does not for the memory-constrained groups. Taken together, these results are the pattern the Platonic Representation Hypothesis predicts, and not the pattern the human-constraint hypothesis predicts: task-scale optimization on its own, without any explicit human-inspired inductive bias, appears to be doing more of the work of aligning a model with human language processing than the biases themselves are.

2 Related Literature

A growing body of work has asked whether LLM representations line up with human brain activity and behavior during language processing. Model-derived features – most often next-word surprisal or hidden-state embeddings – have been shown to predict fMRI and MEG responses during naturalistic listening and reading (Schrimpf et al., 2021; Caucheteux & King, 2022; Caucheteux et al., 2023), to explain variance in fMRI/ECoG recordings beyond what simpler linguistic features capture (Toneva & Wehbe, 2019), and to track a hierarchy of predictive signals across the cortical language network (Caucheteux et al., 2023). These correlations hold up even for models trained on child-scale corpora (Hosseini et al., 2024) and can be strengthened through instruction tuning (Aw et al., 2024), and recent work has found that recurrent architectures match or exceed Transformers on several of these comprehension metrics despite their very different implementation (Michaelov et al., 2024a). A parallel line of work has probed the internal computations of LLMs directly and reported that intermediate layers already encode next-token distributions in a way that resembles human online processing (Kuribayashi et al., 2025), and that reading times can be decoded from hidden states even when the model was never trained to predict them (Tsipidi et al., 2026). Taken together, these findings motivate the Platonic Representation Hypothesis (Huh et al., 2024): the idea that models trained on different objectives, architectures, and data eventually settle into overlapping representational geometry simply because they are all being pushed to solve related tasks over similar sensory and linguistic statistics.

If shared task pressure is sufficient to produce convergence, why constrain a model at all? One answer, pursued mostly in psycholinguistics, is that human sentence processing operates under resource limits that standard LLMs do not share, and that building those limits in should make a model behave more like a human reader even if it does not change its raw task performance. On the architecture side, this has taken the form of restricting the attention mechanism itself: single-head or otherwise capacity-limited attention (Timkey & Linzen, 2023), distance-based decay that biases attention toward nearby tokens (de Varda & Marelli, 2024; Clark et al., 2025; Thamma & Heilbron, 2025), content-sensitive forget gates (Lin et al., 2025) or stick-breaking attention (Tan et al., 2025), and various memory-bounded or resource-rational formulations of the encoding process (Hahn et al., 2022; Lieder & Griffiths, 2019; Xu et al., 2026; Xu & Futrell, 2026; Rasmussen & Schuler, 2018; Keshev et al., 2025). These architectural choices are usually motivated by classic capacity-limited theories of sentence processing, such as activation-based memory retrieval (Lewis & Vasishth, 2005) and lossy-context surprisal (Futrell et al., 2020). On the data side, the BabyLM Challenge takes the opposite route and leaves the architecture alone but restricts training exposure to the 10-100 million words a child hears by a given age, well below the token budgets of standard LLMs (Warstadt et al., 2023a;b; Hu et al., 2024), building on earlier child-directed-language pretraining work (Huebner et al., 2021). Whether either of these constraints actually buys anything in terms of fit to human data is contested: some studies report that smaller, more human-scale models fit reading times better (Wilcox et al., 2024; Oh & Schuler, 2023a), while others find the opposite, or find that fit to human data and standard language-modeling quality trade off in ways that are not well understood (Oh & Schuler, 2023b; Kuribayashi et al., 2021).

A separate and more technical literature asks how, exactly, one should use an LLM's internal state to predict a given human measurement, largely independent of whether the model itself is human-like. Surprisal remains the dominant predictor of reading times and self-paced reading data (Goodkind & Bicknell, 2018; Wilcox et al., 2020; 2023), with psychometric predictive power now benchmarked systematically across model families and sizes (Kuribayashi et al., 2024). Because surprisal is normally read off a model's final output layer, several groups have instead probed intermediate layers directly – via the logit lens (nostalgebraist, 2020) or the tuned lens (Belrose et al., 2023) – to ask which stage of processing best matches human timing (Tenney et al., 2019); Kuribayashi et al. (2025) report that this intermediate-layer surprisal tracks human reading times increasingly well with depth on several behavioral datasets, well before the final layer. Related work has generalized what counts as an "anticipation" signal in online processing more broadly (Giulianelli et al., 2024; Opedal et al., 2024).

Closely related and concurrent with our own work, de Varda et al. (2026) compare a low-dimensional effort predictor (surprisal, frequency, and word length) against high-dimensional final-layer embeddings across eight behavioral and five neural datasets, several of which overlap with ours (Natural Stories, Provo, MECO, ZuCo, Tuckute, UCL). They report that behavioral measures are well ex-

plained by the low-dimensional effort predictor alone, with embeddings adding little, whereas brain responses require the richer embedding space, and that this split coincides with opposite scaling trends within the GPT-2 family – smaller models fit behavior better, larger models fit the brain better. Their question is largely orthogonal to ours: they hold the model class fixed (a single GPT-2 family) and ask which *predictor* – effort features or embeddings – best explains each modality, whereas we hold the predictor fixed (surprisal or full embeddings, applied identically to every model) and ask which *model class* – architecturally or developmentally constrained versus standard – best explains the data, across a substantially larger sweep of models, model families, and neural/behavioral datasets.

Our study differs from this literature along two axes at once. First, rather than picking a side in the constrained-versus-standard debate, we treat it as an empirical question and put memory-constrained architectures, BabyLM-trained models, and standard off-the-shelf LLMs through the exact same evaluation pipeline, so that any advantage for the "human-like" models has to show up as a direct, apples-to-apples effect rather than being read off separate papers with separate datasets and separate metrics. Second, we do this at a scale that, to our knowledge, has not been attempted for this specific comparison: rather than one behavioral corpus or one neuroimaging dataset, we pool self-paced reading and eye-tracking corpora together with fMRI, EEG/ERP, and intracranial recordings, so that a claim of algorithmic convergence (or its absence) is not an artifact of any single modality, measure, or population.

3 Experimental Setup

3.1 Datasets

We assembled a large collection of publicly available corpora spanning two broad categories of human language processing measurements: behavioral and neural. Every dataset is reduced to the same underlying unit of analysis – a word, embedded in its preceding sentence or discourse context – paired with one or more measured human responses to that word. This common representation lets us route any of the model groups described in Section 3.2 through any dataset without dataset-specific modeling code, and it is what allows us to aggregate findings across the datasets in Table 4 (Appendix C) rather than reporting them one at a time.

Behavioral Datasets Behavioral measures index processing difficulty indirectly, through the time it takes a reader to move past a word.

- **Self-paced reading (SPR).** Participants reveal one word (or region) at a time by pressing a button, and reading time is the interval between successive presses. We use the Natural Stories corpus (Futrell et al., 2021), a set of naturalistic narratives deliberately written to contain rare syntactic constructions, and the UCL Reading corpus (Frank et al., 2013), 361 sentences read by 117 participants that we also use as the reference implementation for our reading-time mixed-effects models (Section 3.3).

- **Eye-tracking.** Participants read naturally while an eye tracker records fixations, giving several complementary measures of processing difficulty at a word: *first-pass* (or gaze) duration, the total time spent on a word before the eyes leave it in either direction; *go-past time*, which additionally counts any regressive fixations back to earlier material before progressing past the word; and *total (or single-fixation) dwell time*, summed over every visit to that word across the whole trial. Depending on what each corpus makes available, we extract one or more of these three measures. We draw on the Ghent Eye-tracking Corpus (GECO), the Provo Corpus (Luke & Christianson, 2018), the English subset of the Multilingual Eye-movement Corpus (MECO; Siegelman et al., 2022), and the OneStop corpus, together covering novel reading, norm-controlled passages, and multi-difficulty text. We additionally use the eye-tracking arm of ZuCo (Hollenstein et al., 2018), in which participants read isolated sentences under three separate task instructions (normal reading, sentiment judgment, and relation judgment), giving us three further behavioral sub-datasets from the same recording sessions.

Neural Datasets Neural measures record processing directly from the brain, at complementary spatial and temporal resolutions.

- **fMRI.** We use the Le Petit Prince corpus, in which participants passively listen to an audiobook narration of *The Little Prince* across nine scanning runs, and the Natural Stories fMRI dataset (Shain et al., 2022), which pairs the same narrative materials as the SPR corpus above with BOLD responses pre-averaged within functionally localized regions of the language and multiple-demand networks. We also include Tuckute et al.'s language-network dataset, which measures BOLD responses to isolated sentences, including sentences that were themselves optimized in closed loop to drive or suppress the language network, and the Pereira dataset (Pereira et al., 2018), in which participants read isolated sentences drawn from two separate experiments.
- **EEG / ERP.** We use Broderick et al.'s continuous-listening audiobook EEG corpus, which reduces to a temporal response function between word-level features and the scalp EEG signal. In parallel, we use event-related potential datasets built around the N400 component – a centro-parietal negativity classically associated with semantic expectancy – from controlled psycholinguistic experiments with manipulated cloze probability: a nine-site replication of the classic article/noun expectancy manipulation (Nieuwland et al., 2018), and a large multi-experiment corpus with continuously varying cloze probability (Szewczyk & Federmeier, 2022).
- **Intracranial recordings.** We use a sentence-reading SEEG dataset with electrodes implanted for clinical monitoring (Fedorenko et al., 2016), and a large single-story ECoG dataset recorded during naturalistic podcast listening (Zada et al., 2025).
- **MEG.** We use the MEG-MASC corpus (Gwilliams et al., 2022), in which participants passively listen to naturalistic stories.

Across this collection, the behavioral corpora total well over a hundred thousand word-level observations, and the neural corpora span five recording modalities (fMRI, MEG, scalp EEG, ERP, and intracranial electrophysiology) and several hundred human participants combined – to our knowledge the widest multi-modal sweep assembled to test algorithmic convergence between LLMs and the human language system.

3.2 MODELS

We evaluate every dataset against a common bank of language models, organized into four groups: models built with an explicit human cognitive constraint baked into their architecture, trained either at web scale or under a human-scale data budget; models built with an explicit human developmental constraint baked into their training data alone; and models with neither constraint, trained the standard way at web scale. Crossing the architectural constraint with the data constraint this way turns what might otherwise look like two separate lines of "human-like" modeling work into a single 2×2 design (architecture $\times$ data), letting us ask whether either constraint helps on its own, whether they compound, or neither does. All models are evaluated as frozen feature extractors, never fine-tuned on any of our behavioral or neural targets. Table 3 (Appendix B) lists every model we evaluate, with its depth, parameter count, and HuggingFace repository.

Memory-Constrained Architectures that restrict how much of the preceding context a model's attention mechanism can access or weight, motivated by capacity-limited accounts of human sentence processing such as activation-based memory retrieval (Lewis & Vasishth, 2005) and lossy-context surprisal (Futrell et al., 2020) – capped attention heads, distance-penalized attention, forget-gate and stick-breaking replacements for softmax attention, hard local n-gram windows, and a small sequential recurrent architecture (full list and citations in Appendix B). Every one of these architectures is trained at the same web scale as the Standard LLMs below (a Pile-derived corpus), isolating the architectural constraint from any data constraint.

Constrained-BabyLM The same memory-constrained architectures above, but trained on the BabyLM 10M/100M-word corpora described next instead of web-scale data – the architecture constraint and the data constraint applied together. Keeping this as its own group, rather than folding it into either Memory-Constrained or BabyLM, is what makes the 2×2 crossing explicit: it isolates

what changes when a human-scale-data budget is added on top of an already human-scale architecture, rather than confounding the two.

BabyLM Standard Transformer architectures trained on roughly the number of words a child hears by early adolescence – ten to a hundred million words, several orders of magnitude below the Standard LLMs' token budgets – following the BabyLM Challenge (Warstadt et al., 2023a;b; Hu et al., 2024). We include submissions and baselines from the BabyLM shared tasks spanning several architectural choices under this shared data budget (Appendix B), which lets us ask whether the training-data constraint alone, independent of any architectural change, moves a model toward human-like predictions.

Standard LLMs Off-the-shelf, web-scale autoregressive language models with no human-inspired constraint on either architecture or training data, spanning both standard Transformer and state-space architectures and roughly two orders of magnitude in parameter count (Appendix B lists all twelve model families). For Pythia and OLMo-2, whose intermediate training checkpoints are publicly released, we additionally evaluate every model at several points along its own training trajectory, separating the effect of *how much* a model has learned from the effect of *how it is built*.

3.3 BehavScore and BrainScore Calculation

For every (model, layer, dataset) triple we compute two complementary, independently normalized scores.

BrainScore: predicting the signal from full hidden states. For each layer ℓ we extract contextual hidden states at every word (mean-pooled over subword tokens, full preceding context included in the forward pass) and fit a ridge regression onto the target signal y – an electrode's high-gamma envelope, a voxel's or region's BOLD time course, a scalp channel's EEG amplitude, or a reading time (a lag-matrix TRF, fit jointly across all lags, where y has a temporal profile rather than one scalar per word). Regularization is chosen by nested cross-validation, and we always evaluate out-of-fold, on held-out stories, sections, or sentences. This gives an out-of-fold correlation $r_\ell = \mathrm{corr}(\hat{y}_\ell, y)$. We normalize by a split-half noise ceiling r_{SB} on y (Appendix A.1), which bounds explained variance rather than r itself (van Bree et al., 2025), so the ceiling on r is $\sqrt{r_{\mathrm{SB}}}$:

$$\mathrm{BrainScore}_\ell = \frac{r_\ell}{\sqrt{r_{\mathrm{SB}}}}.$$

We take a model's *last* layer as its primary summary score, $\mathrm{BrainScore} = \mathrm{BrainScore}_L$: since our groups differ enormously in depth (two-layer memory-constrained architectures to 65-layer standard LLMs), reporting $\max_\ell \mathrm{BrainScore}_\ell$ on the same held-out data used for evaluation would let deeper models post a high score by chance alone. We additionally report $\max_\ell \mathrm{BrainScore}_\ell$ as a secondary, exploratory upper bound (Appendix G), but it is not the basis for any headline comparison.

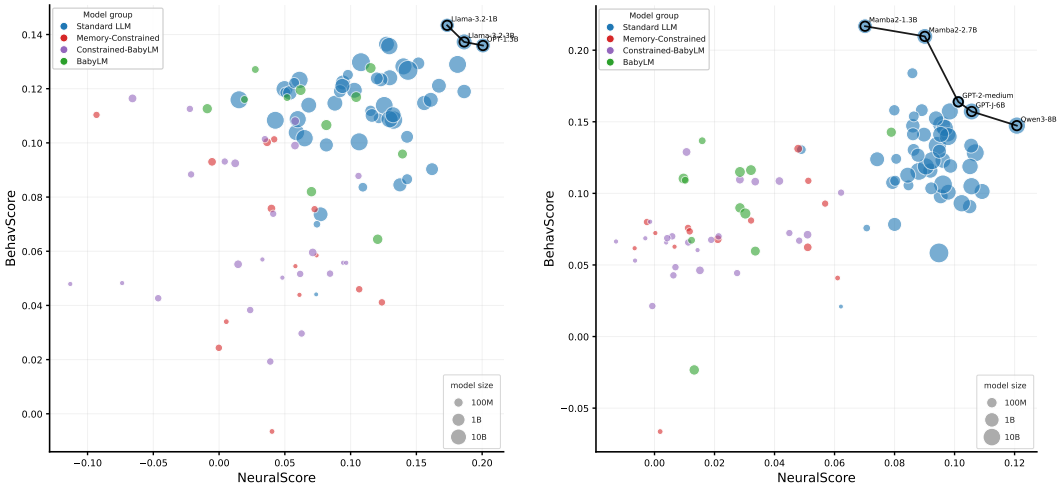
BehavScore: isolating the unique contribution of surprisal. At every layer we obtain a surprisal value via a logit-lens projection (nostalgebraist, 2020) – applying a model's own unembedding matrix to that layer's hidden state – following evidence that intermediate layers already carry human-like predictive information (Kuribayashi et al., 2025). We fit a linear mixed-effects model

$$\texttt{signal} \sim \texttt{wlen} + \texttt{freq} + \underbrace{\texttt{surprisal}_\ell + \texttt{surprisal}_{\ell,\mathrm{prev}}}_{\text{full model only}} + (\text{random effects}),$$

with the dataset-specific random-effects structure detailed in Appendix E, and compare the full model against the baseline that omits the bracketed surprisal terms via the out-of-sample log-likelihood improvement $\Delta LL_\ell = LL_\ell^{\mathrm{full}} - LL^{\mathrm{base}}$. We normalize by a split-half filler-model ceiling $\Delta LL_{\mathrm{ceil}}$ (Appendix A.2), the ΔLL achievable by the best possible single per-item predictor given the data's own reliability (Timkey et al., 2026):

$$\mathrm{BehavScore}_\ell = \frac{\Delta LL_\ell}{\Delta LL_{\mathrm{ceil}}}, \qquad \mathrm{BehavScore} = \mathrm{BehavScore}_L.$$

A model whose surprisal buys no log-likelihood over the baseline scores 0 by this measure, regardless of how large or otherwise capable it is.



(A) All words. **(B)** High DLT-cost words only.

Figure 2: Neural-behavioral trade-off (NeuralScore vs. BehavScore, last layer, surprisal features, both axes DLL-ceiling-normalized) across all evaluated models. **(A)** All words. **(B)** Words restricted to high dependency-locality integration cost (DLT $\geq$ 2, Appendix F), isolating models' behavior specifically on the hardest cases for incremental parsing. Point size is model parameter count; color is model group (Appendix B); the black-outlined path traces the Pareto-optimal frontier.

BrainScore and BehavScore separate two logically distinct claims: BrainScore is how much of a signal a model's full representation can account for; BehavScore is whether the model's own notion of surprise, specifically, is doing any of that work once word length and frequency are controlled for. We report both throughout, and, in Section 4, break each down by model group, model scale, and dataset modality.

4 Results

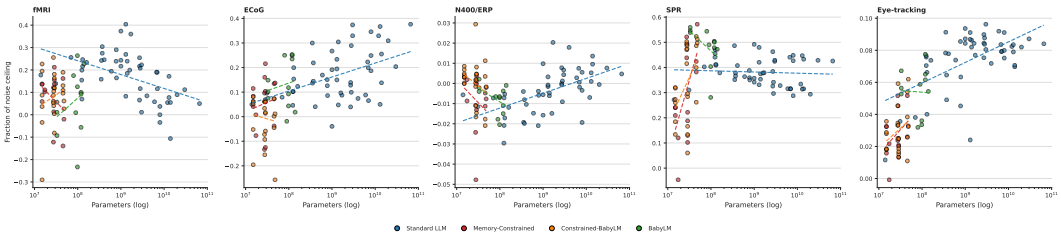
4.1 Surprisal predicts human behavior and neural activity

We first ask how well each group's word-by-word surprisal predicts human data, at the last layer, before turning to embeddings.

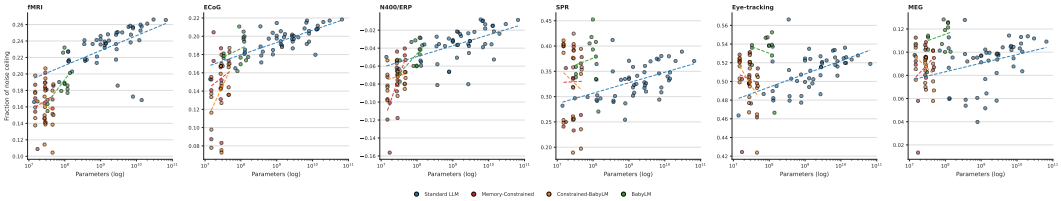
Behavioral (BehavScore). Standard LLMs post the highest mean surprisal-track BehavScore of any group (0.113), with BabyLM close behind (0.108); both memory-constrained groups trail noticeably (Constrained-BabyLM 0.068, Memory-Constrained 0.061). Standard LLM and BabyLM are close enough here that we do not read this as a clear win for either group – what is clear is that both groups without the attention-bounding constraint separate from both groups with it.

Neural (NeuralScore). The ordering is the same but the gaps are larger: Standard LLM leads (0.112), followed by BabyLM (0.071), then Memory-Constrained (0.040) and Constrained-BabyLM (0.025). Unlike on BehavScore, Standard LLM's advantage over BabyLM here is substantial, not marginal.

Figure 2 plots both scores against each other directly. On the full set of words (Figure 2A) the Pareto front is occupied exclusively by three Standard LLMs – Llama-3.2-1B, Llama-3.2-3B, and OPT-1.3B – out of the 100 models with both scores available; no memory-constrained, data-constrained, or architecture-and-data-constrained model reaches the frontier at any point along it. Restricting to the syntactically hardest words (DLT integration cost $\geq$ 2; Figure 2B and Appendix F) does not change this qualitative picture, although it does change which Standard LLMs make up the frontier: Mamba2-1.3B and Mamba2-2.7B anchor the high-BehavScore end, with GPT-2-medium, GPT-J-6B, and Qwen3-8B filling out the rest of the curve toward higher NeuralScore. That the frontier survives intact under a subset chosen specifically to stress incremental-parsing difficulty is,



(A) Surprisal features.



(B) Embedding features.

Figure 3: BehavScore/BrainScore as a function of model parameter count (log scale), by dataset modality and model group, last layer. **(A)** Surprisal features (DLL-ceiling-normalized). **(B)** Embedding features, expressed as fraction of correlation noise ceiling. Dashed lines are per-group linear fits in log-parameter space.

if anything, the harder test for our benchmark's central hypothesis: a psycholinguistically motivated architectural constraint, tested precisely where its motivating theory predicts it should matter most, still does not produce a single model on the Pareto front.

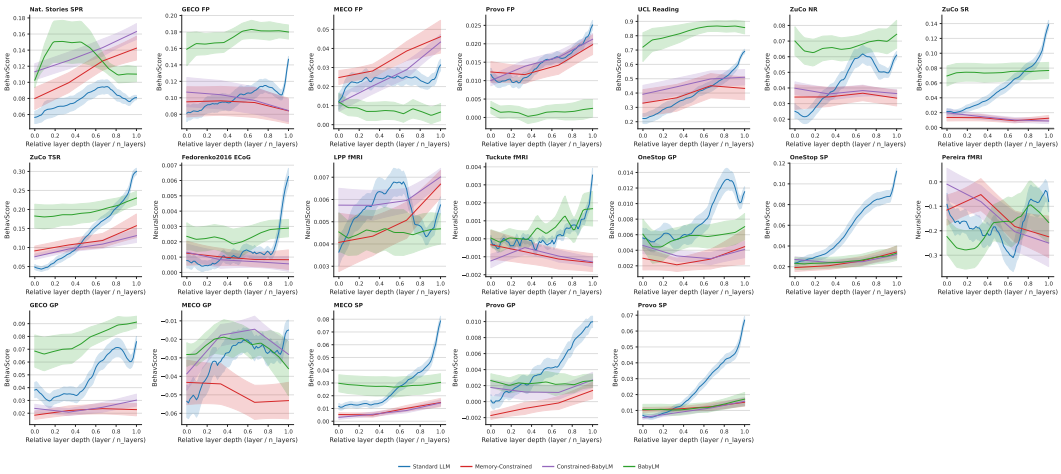
4.2 Scaling with model size

Within the Standard LLM group, larger models are, on the whole, better predictors of both human neural and behavioral responses to surprisal. A log-linear fit of surprisal-track score against parameter count is positive in four of our five dataset modalities (fMRI: $+0.051$; ECoG: $+0.065$; self-paced reading: $+0.005$; eye-tracking: $+0.018$, score units per decade of parameters), with only the N400/ERP modality essentially flat ($+0.003$) – unsurprising given that ERP amplitude is among the noisiest and lowest-N of our signals. Pooling across all eleven surprisal-track datasets that have a computable noise ceiling (Figure 3A), mean DLL-normalized score for Standard LLMs rises from 0.135 in our smallest bin (14–125M parameters) to roughly 0.20–0.24 across bins spanning 1B to 66B parameters; we did not test whether this trend continues, plateaus, or reverses beyond the range of models we evaluate, and do not draw a conclusion about its shape at the largest sizes. The embedding track scales more modestly across the same range but in the same direction in every modality.

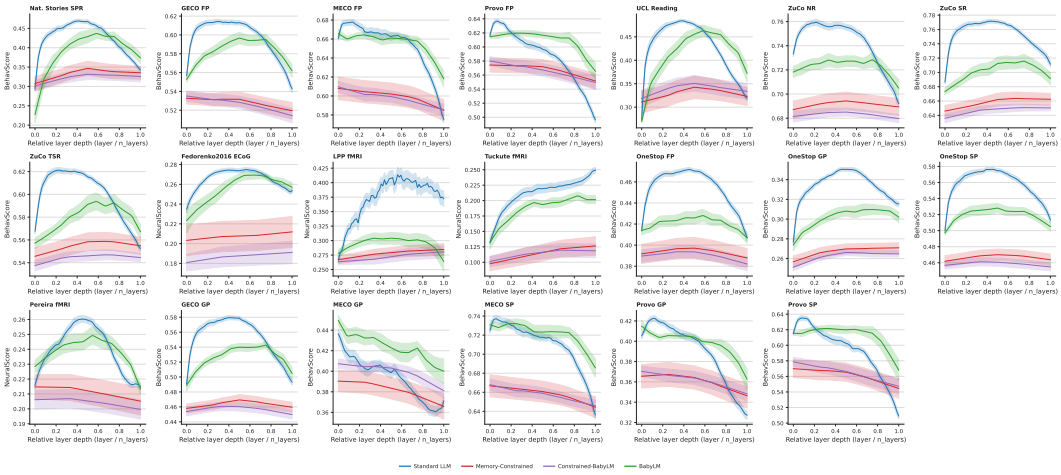
The two memory-constrained groups do not show this pattern as reliably. Their own within-group scaling slopes are noisier and in some modalities negative – Constrained-BabyLM's surprisal-track slope is negative in both fMRI (-0.057) and ECoG (-0.079), and Memory-Constrained is negative in fMRI as well (-0.044) – though we note both groups span a narrower and lower parameter range than the Standard LLMs (roughly 14M to 1.3B, versus up to 66B), so these local slopes are estimated over much less dynamic range and should be read with that caveat.

4.3 Embeddings

The embedding track tells a more qualified version of the same story. On NeuralScore, the ordering matches the surprisal track: Standard LLM leads (0.130), followed by BabyLM (0.105), with Memory-Constrained and Constrained-BabyLM again trailing (0.079 and 0.077). On BehavScore, however, the four groups are close together (0.478–0.512), and BabyLM's mean is nominally the highest of the four – a genuine difference from the surprisal track, where Standard LLM leads on both scores, but one confined to a narrow range around half of the correlation noise ceiling rather than a clear separation between groups. We do not read embedding BehavScore as evidence that



(A) Surprisal features.



(B) Embedding features.

Figure 4: BehavScore/BrainScore as a function of relative layer depth (layer index / total layers), per dataset and model group. **(A)** Surprisal features (DLL-ceiling-normalized). **(B)** Embedding features. Shaded bands are ± 1 SE across models within a group.

any constrained group outperforms Standard LLMs overall: NeuralScore, on the same embeddings, shows the opposite ordering, and a linear ridge regression's fit is also more sensitive than a scalar surprisal predictor to a model's raw representational capacity (Appendix G), which is part of why we treat the surprisal track, not the embedding track, as our primary comparison across groups (Section 3.3).

4.4 Layer depth and layer selection

The two tracks look different across depth. On the surprisal track (Figure 4A), most datasets show scores still rising, or at least not yet declining, at the final layer, for Standard LLM and BabyLM alike. On the embedding track (Figure 4B), the pattern is instead a rise through early-to-middle depth followed by a decline toward the final layer, and this shape is common to all four groups rather than specific to any one of them – a genuinely different depth profile from the surprisal track, not just a weaker version of it.

This motivates checking how much a last-layer summary leaves on the table relative to each model's own best layer. Averaged over all models, last-layer BehavScore recovers a large majority of best-layer BehavScore in every group and on both tracks (85–92% on surprisal, 82–97% on embeddings).

NeuralScore recovers a smaller and more variable share: on the surprisal track this ranges from 18% (Constrained-BabyLM) to 80% (BabyLM), with Standard LLM at 56%; on the embedding track, Standard LLM has the *lowest* recovery fraction of the four groups (47%, versus 49–64% for the other three) – consistent with the decline visible in Figure 4B being, if anything, proportionally steepest for Standard LLM on this track. We do not have a settled explanation for this asymmetry (a natural candidate is that Standard LLMs simply have more layers over which a peak-then-decline profile can unfold, but we have not tested this directly). We report best-layer NeuralScore and BehavScore in full in Appendix G as a secondary, exploratory upper bound; the ordering between groups established above is unchanged there, but for the reasons just given we do not treat it as the basis for any headline claim (Section 3.3).

5 Discussion

We asked whether building human cognitive or developmental constraints into a language model's architecture or training data brings it algorithmically closer to the human language system, or whether standard, unconstrained LLMs already achieve as much algorithmic convergence through task-scale optimization alone. Across seventeen datasets spanning five neural recording modalities and three behavioral measures, we find consistently for the latter. Standard LLMs match or exceed every constrained group on both BrainScore and BehavScore; the Pareto front trading off neural against behavioral predictive power is occupied exclusively by Standard LLMs, whether we look at all words or specifically at the syntactically hardest ones; and scale itself keeps paying off within the unconstrained group in a way it does not for the memory-constrained groups. This is the pattern the Platonic Representation Hypothesis predicts (Huh et al., 2024), and not the pattern the human-constraint hypothesis predicts: shared task demands, not shared inductive biases, appear to be what pulls these systems toward common representations.

The clearest result in support of that reading is architectural. Restricting a Transformer's attention to nearby context – the intervention most directly motivated by a specific human processing theory, capacity-limited working memory (Lewis & Vasishth, 2005; Futrell et al., 2020) – makes a model a *worse* predictor of human neural and behavioral responses, at any training scale. If bounded working memory captured something about human processing that unconstrained models miss, we would expect the opposite. Restricting only the training data, while leaving the architecture untouched (BabyLM), fares better: competitive with Standard LLMs on BehavScore, and briefly ahead of them at the last layer specifically (Appendix G), though never on NeuralScore. The two constraints do not compound – Constrained-BabyLM is not better than either constraint alone – and the constraint with an actual mechanistic motivation is the one that hurts most.

Two design choices shape how we read these results. We did not match model groups on parameter count or perplexity, because the question we ask is whether a constraint helps a model that is otherwise free to be as capable as it can, not whether it helps at fixed capability – a smaller, worse-at-next-token-prediction memory-constrained model is exactly the comparison the working-memory hypothesis needs to win, and it does not. And we treat surprisal, not embeddings, as the primary comparison, because a scalar surprisal value carries no dimensionality advantage for larger models the way a wider hidden state does for ridge regression; that the Standard LLM advantage holds this cleanly on BehavScore is, if anything, the more decisive of our two results. We also note that our benchmark tests predictive power, not internal mechanism – a necessary but not sufficient condition for algorithmic convergence.

Taken together, we read these results as pointing away from architectural constraints motivated by isolated psycholinguistic phenomena, and toward understanding what it is about task-scale optimization itself that produces human-like solutions as a byproduct. If bounded attention, the most theoretically motivated constraint we test, does not help, the more promising path may not be constraining a model's computation at all, but identifying which properties of standard training at scale are actually responsible for the convergence we observe.

Author Contributions

If you'd like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

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A Normalization of BrainScore and BehavScore

Raw correlations and log-likelihood improvements are not comparable across datasets: a Pearson r of 0.3 might be close to ceiling in a noisy single-trial iEEG recording and far from ceiling in a clean, heavily-averaged fMRI dataset. We therefore normalize every score by an estimate of how much signal is actually there to be predicted, using two different procedures for our two metrics, since each requires a different noise ceiling.

A.1 Correlation ceilings: between-subject split-half reliability

For every dataset where a model's score is a correlation between predicted and observed responses (Pearson or Spearman r, at the level of an electrode, voxel, ROI, or channel), we estimate a noise ceiling from the split-half reliability of the human data, following the correction described by van Bree et al. (2025).

Our datasets typically present each item once to each participant, so there is no within-subject repetition to split on. We instead split *across* subjects: for a given dataset and response unit, we repeatedly draw two random, equally sized halves of the subject pool, average each half's response per item, and correlate the two half-averages across items to obtain a split-half correlation r_{sh} – the group-average variant of split-half reliability that van Bree et al. describe for designs without stimulus repetitions. We apply the standard Spearman-Brown correction to extrapolate from a half-sample to the full sample,

$$r_{\text{SB}} = \frac{2\, r_{\text{sh}}}{1 + r_{\text{sh}}},$$

and average r_{SB} over many random subject splits to stabilize the estimate.

It is tempting to treat r_{SB} itself as the ceiling for a model's correlation with the data, but this is the mistake van Bree et al. warn against: r_{SB} is a correlation between two noisy halves, so by the classical correction-for-attenuation logic it bounds *explained variance* (R^2), not the correlation coefficient itself. A model that perfectly recovers the true signal is not itself noisy, so its correlation with the (still noisy) data is attenuated only once rather than twice, giving

$$r_{\max} = \sqrt{r_{\text{SB}}}.$$

We therefore compare every model correlation against $\sqrt{r_{\text{SB}}}$, and every model R^2 against r_{SB} directly, reporting a model's normalized score as the fraction of this ceiling it recovers ($r_{\text{model}} / \sqrt{r_{\text{SB}}}$).

Using r_{SB} as a ceiling for a raw correlation, as roughly sixty percent of the papers surveyed by van Bree et al. do, would systematically understate how far a model actually is from ceiling, especially at the low-to-intermediate reliability that characterizes most of our neural datasets. Ceilings are computed separately per response unit, using the same correlation family – Pearson or Spearman – as the corresponding model score, so a score and its ceiling always live on the same scale.

A.2 Delta log-likelihood ceilings: the split-half filler-model design

For BehavScore, where the quantity of interest is the log-likelihood improvement (ΔLL) a layer's surprisal contributes over a covariate-only baseline, a correlation-based ceiling does not directly apply. We instead adapt the split-half procedure Timkey et al. (2026) use to contextualize their own filler-sentence regression models (their Appendix, Section 3B, "Filler model goodness-of-fit"): we ask how much ΔLL a predictor could achieve in the best possible case, given only the same information available to a single word-level scalar like surprisal – one value per item, no participant- or trial-level access.

We construct this best-case predictor directly from the data. For a given dataset, we randomly split the subject pool into two equal halves, A and B. Within A, we compute the per-item mean response (averaged across subjects and, where relevant, electrodes), substitute it for the surprisal predictor in a mixed-effects model otherwise identical to our real full model, and fit it to the held-out half B. Comparing this model's log-likelihood against a baseline model (identical formula, predictor omitted) fit to the same B data gives one estimate of the achievable ΔLL. We swap the halves – using B's per-item means to predict A – and sum the ΔLL from both directions, repeating over 15 random splits and reporting the mean per-observation ΔLL as the ceiling.

Because the substitute predictor is an empirical group average rather than a noise-free oracle, this ceiling already absorbs the same between-subject averaging noise a real surprisal predictor faces, while staying agnostic to any particular theory of processing difficulty – it is simply the most any single per-item scalar could explain, given the data's own reliability. We report a model's raw ΔLL alongside this ceiling and, where useful, as a percentage of it, exactly as Timkey et al. report their filler models' fit in absolute and relative terms. Two details carry over from our main fitting procedure: ceilings are computed out-of-sample, using the same item/chunk train-test split as the corresponding real analysis, and the ceiling model's random-effects structure mirrors the real model's for that dataset – so the only thing distinguishing the two is the substitution of an empirical per-item mean for the model-derived surprisal.

A.3 Reported ceilings

For transparency, Tables 1 and 2 report every noise ceiling actually used to normalize the scores in this paper, computed by the two procedures above, including datasets whose ceiling is comparatively low. A low ceiling indicates that the human measurement itself carries limited reliable signal to recover, and a normalized score built on it should be interpreted with that in mind regardless of how favorable the resulting ratio appears.

Table 1: Delta log-likelihood noise ceilings (BehavScore normalization), computed via the split-half item-mean-as-predictor procedure of Appendix A.2, following Timkey et al. (2026). Ceiling is per-observation ΔLL; "splits conv." reports how many of 10–30 random subject-half splits converged. All 23 dataset/measure entries used for BehavScore in this paper return a positive ceiling; several small-N entries carry higher relative noise but no split failed to converge.

Dataset	Ceiling	±SD	Rel. SD	Splits conv.	Note
Fedorenko2016 (ECoG)	0.00832	0.00162	19%	10/10	small N (5 subj.)
GECO (FP)	0.00866	0.00036	4%	10/10	small N (14 subj.)
GECO (GP)	0.00416	0.00043	10%	15/15	small N (14 subj.)
Le Petit Prince (fMRI)	0.01015	0.00088	9%	15/15	
MECO (FP)	0.02336	0.00368	16%	15/15	
MECO (GP)	0.00797	0.00186	23%	15/15	
MECO (SP)	0.03786	0.00256	7%	15/15	

Table 1 – continued

Dataset	Ceiling	±SD	Rel. SD	Splits conv.	Note
Natural Stories (SPR)	0.02381	0.00091	4%	15/15	
Natural Stories (fMRI)	0.02966	0.00394	13%	15/15	
Nieuwland (N400)	0.01250	0.00120	10%	30/30	
OneStop (FP)	0.01286	0.00056	4%	15/15	
OneStop (GP)	0.01429	0.00089	6%	15/15	
OneStop (SP)	0.02359	0.00139	6%	15/15	
Pereira (fMRI)	0.00288	0.00233	81%	15/15	small N ($\leq$10 subj.)
Podcast (ECoG)	0.00811	0.00101	12%	10/10	small N (9 subj.)
Provo (FP)	0.03354	0.00150	4%	15/15	
Provo (GP)	0.03596	0.00160	4%	15/15	
Provo (SP)	0.04573	0.00102	2%	15/15	
Tuckute (fMRI)	0.00952	0.00132	14%	10/10	
UCL Reading (SPR)	0.00236	0.00026	11%	15/15	
ZuCo Reading (NR)	0.02782	0.00225	8%	10/10	small N (12 subj.)
ZuCo Reading (SR)	0.03143	0.00155	5%	15/15	small N (12 subj.)
ZuCo Reading (TSR)	0.01248	0.00239	19%	15/15	small N (12 subj.)

Table 2: Split-half correlation noise ceilings (BrainScore normalization, $\sqrt{r_{\mathrm{SB}}}$, Appendix A.1) for every dataset in the embedding/ridge track, and the fraction of that ceiling reached by the best model. Rows marked † have a ceiling below 0.01 – an absolute reliability so low that a high percent-of-ceiling figure reflects very little recoverable signal in the human data itself, not a strong model fit; we report these for completeness but do not treat them as headline results.

Dataset	Modality	Ceiling ($\sqrt{r_{\mathrm{SB}}}$)	Best model's % of ceiling
ZuCo Reading SR	Behavioral	0.819	81%
ZuCo Reading	Behavioral	0.819	79%
MECO SP	Behavioral	0.930	79%
MECO	Behavioral	0.899	72%
Provo SP	Behavioral	0.943	69%
Provo	Behavioral	0.933	68%
GECO	Behavioral	0.627	65%
ZuCo Reading TSR	Behavioral	0.819	65%
GECO GP	Behavioral	0.529	62%
OneStop SP	Behavioral	0.888	62%
UCL Reading	Behavioral	0.689	56%
Natural Stories SPR	Behavioral	0.914	55%
MECO GP	Behavioral	0.684	51%
OneStop	Behavioral	0.829	50%
Le Petit Prince fMRI	fMRI	0.912	49%
Provo GP	Behavioral	0.919	47%
Fedorenko2016 ECoG	ECoG	0.711	40%
OneStop GP	Behavioral	0.776	38%
Nieuwland N400 EEG	EEG (N400)	0.344	35%
Tuckute fMRI	fMRI	0.480	34%
Pereira fMRI	fMRI	0.507	30%
Natural Stories fMRI	fMRI	0.628	28%
Podcast iEEG	iEEG	0.398	23%
MEG-MASC (Gwilliams)	MEG	0.097	20%

B Full List of Evaluated Models

Table 3 lists every model evaluated in this benchmark, with its family, transformer/SSM depth, parameter count, and HuggingFace repository, corresponding to the four model groups introduced in Section 3.2.

The Memory-Constrained and Constrained-BabyLM groups (identical architectures, web-scale vs. BabyLM-scale training data) cover five families: architectures that cap the number of attention heads available for long-range dependencies (Timkey & Linzen, 2023); architectures that bias attention toward nearby tokens through a learned or fixed distance penalty, including ALiBi (Press et al., 2021; de Varda & Marelli, 2024; Clark et al., 2025); architectures that replace softmax attention with a content-sensitive forget gate (Lin et al., 2025) or a stick-breaking process (Tan et al., 2025); hard local n-gram windows that can only ever attend to the two, three, or five preceding tokens regardless of what the input contains (Bu et al., 2026); and CBR-RNN (and its variant CBR-RNN-M), a small recurrent architecture that processes context sequentially rather than through attention at all (Timkey & Linzen, 2023). These architectures and their BabyLM-scale-trained counterparts were obtained from the public HuggingFace checkpoints released alongside Bu et al. (2026). The BabyLM group spans submissions and baselines from the BabyLM shared tasks – including the organizers' own baselines (Charpentier et al., 2025), a masked-language-modeling variant (Edman & Fraser, 2025), an instruction-tuned small model (Capone et al., 2025), a mixture-of-experts architecture (Tapaninaho, 2025), and a model trained with interactive storytelling feedback (Mayer Martins et al., 2025) – alongside the earlier two BabyLM challenges (Warstadt et al., 2023a;b; Hu et al., 2024), spanning causal Transformer baselines, mixture-of-experts variants, and hybrid causal/masked objectives, all likewise obtained from their public HuggingFace repositories. The Standard LLM group spans twelve off-the-shelf families with no human-inspired constraint on either architecture or training data: GPT-2 (Radford et al., 2019), GPT-Neo (Black et al., 2021), GPT-J (Wang & Komatsuzaki, 2021), GPT-NeoX (Black et al., 2022), OPT (Zhang et al., 2022), Mamba and Mamba-2 (Gu & Dao, 2024), Pythia (Biderman et al., 2023), OLMo-2 (Team OLMo et al., 2025), Qwen3 (Yang et al., 2025), Llama-3.2, and Ministral, from around 100 million to tens of billions of parameters and including both standard Transformer and state-space architectures.

Table 3: All models evaluated in this benchmark, grouped by family within the three model classes of Section 3.2 (rules separate each fine-grained family). Parameter and layer counts follow the originating release (or the loaded config for the memory-constrained and BabyLM checkpoints; see `neuroai/models/registry.py`). Training-token/word counts are reported only where documented with confidence by the originating source; a blank cell means this was not reliably available rather than zero. For Pythia and OLMo-2, the listed repository is evaluated both at its final (`main`) checkpoint and at multiple intermediate training steps (Section 3.2).

Family	HuggingFace model ID	#Params	#Layers	Tr. tokens	Ref.
Standard LLMs					
GPT-2	openai-community/gpt2	124M	13		(Radford et al., 2019)
GPT-2	openai-community/gpt2-medium	355M	25		(Radford et al., 2019)
GPT-2	openai-community/gpt2-large	774M	37		(Radford et al., 2019)
GPT-2	openai-community/gpt2-xl	1.6B	49		(Radford et al., 2019)
GPT-Neo/J/NeoX	EleutherAI/gpt-neo-125m	125M	13		(Black et al., 2021)
GPT-Neo/J/NeoX	EleutherAI/gpt-neo-1.3B	1.3B	25		(Black et al., 2021)
GPT-Neo/J/NeoX	EleutherAI/gpt-neo-2.7B	2.7B	33		(Black et al., 2021)
GPT-Neo/J/NeoX	EleutherAI/gpt-j-6b	6B	29	402B	(Wang & Komatsuzaki, 2021)

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Table 3 – continued

Family	HuggingFace model ID	#Params	#Layers	Tr. tokens	Ref.
GPT-Neo/J/NeoX	EleutherAI/gpt-neox-20b	20B	45	472B	(Black et al., 2022)
OPT	facebook/opt-125m	125M	13	180B	(Zhang et al., 2022)
OPT	facebook/opt-350m	350M	25	180B	(Zhang et al., 2022)
OPT	facebook/opt-1.3b	1.3B	25	180B	(Zhang et al., 2022)
OPT	facebook/opt-2.7b	2.7B	33	180B	(Zhang et al., 2022)
OPT	facebook/opt-6.7b	6.7B	33	180B	(Zhang et al., 2022)
OPT	facebook/opt-13b	13B	41	180B	(Zhang et al., 2022)
OPT	facebook/opt-30b	30B	49	180B	(Zhang et al., 2022)
OPT	facebook/opt-66b	66B	65	180B	(Zhang et al., 2022)
Mamba	state-spaces/mamba-130m-hf	130M	25	300B	(Gu & Dao, 2024)
Mamba	state-spaces/mamba-370m-hf	370M	49	300B	(Gu & Dao, 2024)
Mamba	state-spaces/mamba-790m-hf	790M	49	300B	(Gu & Dao, 2024)
Mamba	state-spaces/mamba-1.4b-hf	1.4B	49	300B	(Gu & Dao, 2024)
Mamba	state-spaces/mamba-2.8b-hf	2.8B	61	300B	(Gu & Dao, 2024)
Mamba-2	AntonV/mamba2-130m-hf	130M	25	300B	(Dao & Gu, 2024)
Mamba-2	AntonV/mamba2-370m-hf	370M	49	300B	(Dao & Gu, 2024)
Mamba-2	AntonV/mamba2-780m-hf	780M	49	300B	(Dao & Gu, 2024)
Mamba-2	AntonV/mamba2-1.3b-hf	1.3B	49	300B	(Dao & Gu, 2024)
Mamba-2	AntonV/mamba2-2.7b-hf	2.7B	61	300B	(Dao & Gu, 2024)
Pythia	EleutherAI/pythia-14m-deduped	14M	6	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-31m-deduped	31M	6	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-70m-deduped	70M	6	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-160m-deduped	160M	12	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-410m-deduped	410M	24	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-1b-deduped	1B	17	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-1.4b-deduped	1.4B	25	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-2.8b-deduped	2.8B	33	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-6.9b-deduped	6.9B	33	300B	(Biderman et al., 2023)
Pythia	EleutherAI/pythia-12b-deduped	12B	37	300B	(Biderman et al., 2023)
OLMo-2	allenai/OLMo-2-0425-1B	1B	17	4.0T	(Team OLMo et al., 2025)
OLMo-2	allenai/OLMo-2-1124-7B	7B	32	3.9T	(Team OLMo et al., 2025)
OLMo-2	allenai/OLMo-2-1124-13B	13B	40	5.0T	(Team OLMo et al., 2025)
Qwen3	Qwen/Qwen3-0.6B-Base	600M	29	36T	(Yang et al., 2025)
Qwen3	Qwen/Qwen3-1.7B-Base	1.7B	29	36T	(Yang et al., 2025)
Qwen3	Qwen/Qwen3-4B-Base	4B	37	36T	(Yang et al., 2025)
Qwen3	Qwen/Qwen3-8B-Base	8B	37	36T	(Yang et al., 2025)
Qwen3	Qwen/Qwen3-14B-Base	14B	41	36T	(Yang et al., 2025)
Llama-3	meta-llama/Llama-3.2-1B	1B	17	9T	
Llama-3	meta-llama/Llama-3.2-3B	3B	29	9T	
Llama-3	meta-llama/Llama-3.1-8B	8B	33	15T	
Ministral	mistralai/Ministral-3-3B-Base-2512	3B	29		
Ministral	mistralai/Ministral-3-8B-Base-2512	8B	37		
Ministral	mistralai/Ministral-3-14B-Base-2512	14B	41		
Memory-Constrained					
CBR-RNN	cclark/cbr-rnn	17.4M	1		(Timkey & Linzen, 2023)
CBR-RNN	cclark/cbr-rnn-m	17.4M	1		(Timkey & Linzen, 2023)
Stick-Breaking	Lanni-ni/stickbreaking_pile_2layer	27.8M	2		(Tan et al., 2025)
Stick-Breaking	Lanni-ni/stickbreaking_pile_4layer	48.1M	4		(Tan et al., 2025)

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Table 3 – continued

Family	HuggingFace model ID	#Params	#Layers	Tr. tokens	Ref.
Stick-Breaking	Lanni-ni/stickbreaking_babylm_10m_21ayer	27.8M	2	∼10M w.	(Tan et al., 2025)
Stick-Breaking	Lanni-ni/stickbreaking_babylm_10m_41ayer	48.1M	4	∼10M w.	(Tan et al., 2025)
Stick-Breaking	Lanni-ni/stickbreaking_babylm_100m_2layer	27.8M	2	∼100M w.	(Tan et al., 2025)
Stick-Breaking	Lanni-ni/stickbreaking_babylm_100m_4layer	48.1M	4	∼100M w.	(Tan et al., 2025)
ALiBi	Lanni-ni/alibi_pile_2layer	27.4M	2		(Press et al., 2021)
ALiBi	Lanni-ni/alibi_pile_4layer	45.7M	4		(Press et al., 2021)
ALiBi	Lanni-ni/alibi_babylm_10m_2layer	27.4M	2	∼10M w.	(Press et al., 2021)
ALiBi	Lanni-ni/alibi_babylm_10m_4layer	45.7M	4	∼10M w.	(Press et al., 2021)
ALiBi	Lanni-ni/alibi_babylm_100m_2layer	27.4M	2	∼100M w.	(Press et al., 2021)
ALiBi	Lanni-ni/alibi_babylm_100m_4layer	45.7M	4	∼100M w.	(Press et al., 2021)
Forgetting-Transf.	Lanni-ni/forgetting_pile_2layer	27.4M	2		(Lin et al., 2025)
Forgetting-Transf.	Lanni-ni/forgetting_pile_4layer	45.7M	4		(Lin et al., 2025)
Forgetting-Transf.	Lanni-ni/forgetting_babylm_10m_2layer	27.4M	2	∼10M w.	(Lin et al., 2025)
Forgetting-Transf.	Lanni-ni/forgetting_babylm_10m_4layer	45.7M	4	∼10M w.	(Lin et al., 2025)
Forgetting-Transf.	Lanni-ni/forgetting_babylm_100m_2layer	27.4M	2	∼100M w.	(Lin et al., 2025)
Forgetting-Transf.	Lanni-ni/forgetting_babylm_100m_4layer	45.7M	4	∼100M w.	(Lin et al., 2025)
Hard 2-gram	Lanni-ni/hard_2gram_pile_2layer	15M	2		(Bu et al., 2026)
Hard 2-gram	Lanni-ni/hard_2gram_pile_4layer	28.8M	4		(Bu et al., 2026)
Hard 2-gram	Lanni-ni/hard_2gram_babylm_10m_2layer	15M	2	∼10M w.	(Bu et al., 2026)
Hard 2-gram	Lanni-ni/hard_2gram_babylm_10m_4layer	28.8M	4	∼10M w.	(Bu et al., 2026)
Hard 2-gram	Lanni-ni/hard_2gram_babylm_100m_2layer	15M	2	∼100M w.	(Bu et al., 2026)
Hard 2-gram	Lanni-ni/hard_2gram_babylm_100m_4layer	28.8M	4	∼100M w.	(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_pile_2layer	15M	2		(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_pile_4layer	28.8M	4		(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_babylm_10m_2layer	15M	2	∼10M w.	(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_babylm_10m_4layer	28.8M	4	∼10M w.	(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_babylm_100m_2layer	15M	2	∼100M w.	(Bu et al., 2026)
Hard 3-gram	Lanni-ni/hard_3gram_babylm_100m_4layer	28.8M	4	∼100M w.	(Bu et al., 2026)
Hard 5-gram	Lanni-ni/hard_5gram_pile_2layer	15M	2		(Bu et al., 2026)
Hard 5-gram	Lanni-ni/hard_5gram_pile_4layer	28.8M	4		(Bu et al., 2026)
Hard 5-gram	Lanni-ni/hard_5gram_babylm_10m_2layer	15M	2	∼10M w.	(Bu et al., 2026)
Hard 5-gram	Lanni-ni/hard_5gram_babylm_10m_4layer	28.8M	4	∼10M w.	(Bu et al., 2026)

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Table 3 – continued

Family	HuggingFace model ID	#Params	#Layers	Tr. tokens	Ref.
Hard 5-gram	Lanni-ni/hard_5gram_babylm_100m_2layer	15M	2	∼100M w.	(Bu et al., 2026)
Hard 5-gram	Lanni-ni/hard_5gram_babylm_100m_4layer	28.8M	4	∼100M w.	(Bu et al., 2026)
BabyLMs					
BabyLM	colinglab/CLASS_IT-140M	140M	13		(Capone et al., 2025)
BabyLM	llm-slice/blm-gpt2s-90M-s42_901M-s42_submission	90M	13		
BabyLM	Jtapsa/moep	80M	13		(Tapaninaho, 2025)
BabyLM	BabyLM-community/BabyLM-2026-Baseline-GPT2-Strict	98.4M	13	∼100M w.	(Charpentier et al., 2025)
BabyLM	BabyLM-community/BabyLM-2026-Baseline-GPT2-Strict-Small	98.4M	13	∼10M w.	(Charpentier et al., 2025)
BabyLM	BabyLM-community/babylm-baseline-100m-gpt-bert-causal-focus	119.5M	13	∼100M w.	(Edman & Fraser, 2025)
BabyLM	BabyLM-community/babylm-baseline-100m-gpt-bert-mixed	119.5M	13	∼100M w.	(Edman & Fraser, 2025)
BabyLM	BabyLM-community/babylm-baseline-100m-gpt-bert-masked-focus	119.5M	13	∼100M w.	(Edman & Fraser, 2025)
BabyLM	BabyLM-community/babylm-baseline-10m-gpt-bert-causal-focus	33M	13	∼10M w.	(Edman & Fraser, 2025)
BabyLM	BabyLM-community/babylm-baseline-10m-gpt-bert-mixed	33M	13	∼10M w.	(Edman & Fraser, 2025)
BabyLM	BabyLM-community/babylm-baseline-10m-gpt-bert-masked-focus	33M	13	∼10M w.	(Edman & Fraser, 2025)

C Full List of Evaluated Datasets

Table 4 lists every dataset evaluated in this benchmark, grouped by modality, corresponding to the behavioral and neural categories introduced in Section 3.1.

Table 4: All datasets evaluated in this benchmark, grouped by modality. N items reports the unit each dataset's items are counted in (sentences, passages, screens, stories, electrodes, ...), since this varies by design. Eye-tracking measures: FP = first-pass/gaze duration, GP = go-past time, SP = (single-fixation or total) dwell time. N subj. reflects each study's usable sample after its own quality-control exclusions.

Dataset	Modality	Subj.	Items	Measure(s)	Ref.
Self-Paced Reading					
Natural Stories SPR	SPR	180	10 stories (10,256 word pos.)	raw RT	(Futrell et al., 2021)
UCL Reading	SPR	117	361 sentences	raw RT	(Frank et al., 2013)
Eye-Tracking					
GECO	Eye-tracking	14	588 screens	FP, GP	(Cop et al., 2017)
MECO (English)	Eye-tracking	46	12 passages	SP, FP, GP	(Siegelman et al., 2022)
Provo	Eye-tracking	84	55 texts	SP, FP, GP	(Luke & Christianson, 2018)

continued on next page

Table 4 – continued

Dataset	Modality	Subj.	Items	Measure(s)	Ref.
OneStop	Eye-tracking	360	330 passages	SP, FP, GP	(Berzak et al., 2025)
ZuCo Reading	Eye-tracking	12	SR/NR/TSR: 400/300/407 sent.	TRT	(Hollenstein et al., 2018)
N400 / ERP					
Nieuwland	EEG (ERP)	334	9-lab RSVP replication	N400	(Nieuwland et al., 2018)
Szewczyk	EEG (ERP)	$\sim$138	$\sim$1,900–2,000 items	N400	(Szewczyk & Federmeier, 2022)
Continuous EEG					
Broderick	EEG	19	20 segs. ($\sim$196s each)	EEG (1–8Hz)	(Broderick et al., 2018)
MEG					
Gwilliams (MEG-MASC)	MEG	27	4 stories $\times$ 2 sess.	continuous MEG	(Gwilliams et al., 2022)
fMRI					
Le Petit Prince	fMRI	49	9 sections (audio)	BOLD	(Li et al., 2022)
Natural Stories fMRI	fMRI	78	8 narratives (26 fROIs)	BOLD	(Shain et al., 2022)
Tuckute	fMRI	10	2,000 sentences (6 ROIs)	BOLD	(Tuckute et al., 2024)
Pereira	fMRI	$\leq$10	627 sentences (12 ROIs)	BOLD	(Pereira et al., 2018)
iEEG / ECoG					
Podcast	ECoG	9	1 story, >5,000 w. (1,330 electr.)	high-gamma	(Zada et al., 2025)
Fedorenko2016	iEEG	5	52 sentences (416 words)	iEEG (scalar)	(Fedorenko et al., 2016)

D Implementation Details

Data provenance. All datasets were obtained from their public OpenNeuro or OSF repositories under the terms specified therein.

Feature extraction. All language models are loaded in bfloat16 and evaluated as frozen feature extractors, with no gradient updates. For each word, hidden states are mean-pooled over its constituent subword tokens at every layer; unless otherwise noted (Section 3.1), a word's representation is computed with its full preceding sentence or discourse context in the forward pass, rather than in isolation. Surprisal at every layer is obtained via a logit-lens projection, applying each model's own unembedding matrix to the corresponding intermediate hidden state.

Ridge regression. For every dataset, nested cross-validation selects the ridge penalty from a log-spaced grid via an inner search, with predictors standardized by a robust (median/IQR) scaler fit on the training fold only, and outer folds grouped so that no sentence, item, story, or run straddles the train/test boundary. The grid and fold count are tuned per dataset to its sample size and predictor dimensionality rather than fixed globally: Table 5 lists the exact configuration for every dataset. We report mean out-of-fold Pearson and Spearman correlation across the outer folds, and sweep this procedure across every layer of every model to obtain a layerwise profile.

Table 5: Ridge regression configuration for each dataset's BrainScore encoder: log-spaced α grid (number of grid points in parentheses), outer cross-validation fold count, and the grouping unit used so that no fold splits correlated observations. GroupKFold groups on the listed unit unless marked LOSO (leave-one-unit-out, i.e. fold count equals unit count). Datasets with no embedding/ridge track (OneStop, Szewczyk) are omitted.

Dataset	α grid (points)	Folds	Grouping unit
GECO (FP/GP)	10^{1}–10^{10} (10)	5	screen
MECO (FP/GP/SP)	$10^{0.5}$–10^{10} (12)	5	passage
Provo (FP/GP/SP)	10^{0}–10^{10} (11)	5	text
UCL Reading	$10^{1.5}$–10^{9} (9)	5	sentence
Natural Stories SPR	10^{-1}–10^{5} (7)	10 (LOSO)	story
Natural Stories fMRI	10^{-2}–10^{4} (6)	8 (LOSO)	story
ZuCo Reading (NR)	$10^{2.5}$–10^{9} (8)	5	sentence
ZuCo Reading (SR/TSR)	10^{2}–10^{6} (8)	5	sentence
Nieuwland	$10^{3.5}$–10^{10} (6)	10	sentence item
Tuckute	10^{2}–10^{7} (5)	4	sentence
Broderick	$10^{3.5}$–10^{8} (5)	3	audiobook segment
Le Petit Prince	10^{3}–$10^{5.3}$ (8)	9 (LOSO)	section
Podcast	$10^{3.5}$–10^{8} (5)	3	contiguous time chunk
Fedorenko2016	$10^{3.5}$–10^{8} (5)	5	sentence
Pereira	10^{1}–10^{6} (8)	5	passage

Temporal response function (TRF) datasets. For the three datasets with a temporal response rather than a single scalar per word – Broderick, Podcast, and Fedorenko2016 – the ridge encoder fits a lag-matrix TRF: one multi-output ridge regression predicts every time lag jointly from the same set of predictors, rather than one regression per lag. Broderick's EEG is downsampled to 32 Hz and epoched from -0.2 to 1.0 s relative to word onset (38 lags); Podcast's ECoG uses the same 32 Hz rate over a wider -1.0 to 1.0 s window (128 lags). Fedorenko2016 has no temporal axis: its intracranial recordings are already reduced to a single scalar response per (word, electrode) pair, so no lag-matrix step is needed there.

Podcast's pooled mixed-effects model. In addition to its per-electrode ridge encoder, Podcast is also fit with a pooled linear mixed-effects model in which electrodes are nested within subject (Appendix E), separately *at each lag* of a -0.2 to 0.8 s, 32-lag window – one model per (lag, LM layer) pair rather than one model for the whole epoch. Held-out partitions are formed from contiguous 30-word chunks of the continuous transcript, the direct word-index analogue of the chunk-based splits used for the fMRI datasets below.

fMRI encoding pipelines. For Le Petit Prince and Natural Stories fMRI, word-level predictors are convolved with a canonical double-gamma HRF (Glover, 1999), implemented via `nilearn`, before ridge or mixed-effects fitting; both use $\mathrm{TR} = 2$ s and discard the first and last several volumes of each run for signal equilibration. Le Petit Prince additionally restricts voxels to the top 25% by split-half reliability before averaging into ten bilateral parcels of the Fedorenko language-network atlas, and both its ridge and mixed-effects analyses are computed on the group-average BOLD signal rather than per subject, following the most-reliable-voxel selection and subject-averaging procedure of Bonnasse-Gahot & Pallier (2024). Natural Stories fMRI aggregates BOLD into 26 functional regions of interest (6 left-hemisphere language-network parcels, 20 bilateral multiple-demand parcels) per Shain et al. (2022). Tuckute's sentence-level design uses only the 1,000-sentence naturalistic condition for encoding and mixed-effects analyses; the two adversarially optimized conditions designed to drive or suppress the language network are evaluated separately, since only 2–3 of the 10 participants saw each of those sentences.

Linear mixed-effects models. All LME models are fit in R via `lme4`. Where out-of-sample evaluation is used, held-out partitions are formed from contiguous chunks of the data (words, sentences,

or short time windows depending on the dataset), shuffled with a fixed seed before assignment to train/held-out splits, so that both partitions sample the full range of each recording or corpus rather than, e.g., only its first half. If a model with the full random-effects structure specified in Appendix E fails to converge, we follow the simplification procedure of Barr et al. (2013): the correlation between random slopes is dropped first, and if convergence still fails, the slope itself is dropped, leaving a random-intercept-only term. This fallback is applied identically across every layer of a given model so that log-likelihood comparisons remain valid.

E Linear Mixed-Effects Formulas

This appendix specifies the linear mixed-effects (LME) models underlying the BehavScore and BrainScore analyses of Section 3.3, in standard `lme4` formula notation. Each model is fit once per transformer layer: a *baseline* model containing only nuisance covariates, and a *full* model that additionally includes the layer's surprisal, with the two compared via the log-likelihood difference described in Appendix A.2. All continuous predictors are z-scored within dataset; `surprisal_z` denotes the per-layer logit-lens surprisal of the evaluated model, and `prev_surprisal_z` the same quantity for the preceding word (a spillover control).

E.1 Self-paced reading and eye-tracking

Reading times and fixation durations across Natural Stories, UCL Reading, GECO, MECO, Provo, OneStop, and ZuCo Reading are modeled with a common specification, with word length, frequency, and sentence position as nuisance covariates and by-subject random slopes:

```
baseline: fdur ~ sentpos_z + wlen_z + freq_z +
                 (1 + sentpos_z + wlen_z | subject)
full: fdur ~ sentpos_z + wlen_z + freq_z + surprisal_z
+ prev_surprisal_z +
             (1 + sentpos_z + wlen_z + surprisal_z +
prev_surprisal_z | subject)
```

E.2 N400 event-related potentials

Nieuwland is modeled following the specification of its originating study rather than our shared self-paced-reading/eye-tracking structure, since its multi-site design calls for a different random-effects treatment. Following Nieuwland et al. (2018), laboratory is entered as a fixed rather than random effect, since a random-effect variance is not reliably estimable from only nine sites; there is no electrode term, as N400 amplitude is already averaged across centro-parietal channels in this dataset, nor a frequency covariate, matching the original analysis:

```
baseline: n400 ~ lab + (1 | subject) + (1 | item)
full: n400 ~ lab + surprisal_z + (1 | subject) + (1 |
item)
```

E.3 Continuous EEG

For Broderick, random effects are estimated separately for each subject and the resulting log-likelihood differences are averaged across subjects, rather than pooling all subjects into a single model. `region` denotes four anatomical scalp quadrants (left, right, posterior, occipital).

```
baseline: eeg ~ wlen_z + freq_z + (1 | region)
full: eeg ~ wlen_z + freq_z + surprisal_z +
prev_surprisal_z + (1 + surprisal_z | region)
```

E.4 fMRI

For Natural Stories fMRI, `story` is a crossed random effect, and no spillover term is included, since HRF convolution already blends the contributions of adjacent words into a single BOLD sample. Omitting the frequency covariate reproduces the specification of Shain et al. (2022).

```
baseline: bold ~ wlen_z + freq_z + sentpos_z +
                  (1 + wlen_z + freq_z | subject) + (1 |
story)
full: bold ~ wlen_z + freq_z + sentpos_z + surprisal_z
+
              (1 + wlen_z + freq_z + surprisal_z |
subject) + (1 + surprisal_z | story)
```

For Le Petit Prince, the model is fit on the average-subject BOLD signal; `region` denotes one of ten bilateral language-network parcels from the Fedorenko functional atlas, and `section` one of the nine narrative chapters.

```
baseline: bold ~ wlen_z + freq_z + sentpos_z + (1 |
region) + (1 | section)
full: bold ~ wlen_z + freq_z + sentpos_z + surprisal_z
+
              (1 + surprisal_z | region) + (1 |
section)
```

For Tuckute, the response is a single BOLD scalar per (subject, ROI, sentence), so word length and frequency are averaged over the sentence (`mean_wlen_z`, `mean_freq_z`) and surprisal is computed over the whole sentence rather than word-by-word. The model is fit on the naturalistic sentence condition only.

```
baseline: y ~ mean_wlen_z + mean_freq_z +
                (1 + mean_wlen_z + mean_freq_z | subject)
+ (1 | roi) + (1 | sentence_id)
full: y ~ mean_wlen_z + mean_freq_z + surprisal_z +
           (1 + mean_wlen_z + mean_freq_z + surprisal_z |
subject) + (1 | roi) +
           (1 + surprisal_z | sentence_id)
```

For Pereira, which uses the same per-sentence design as Tuckute but with sentences of variable length, we follow the specification of Lin & Schuler (2026) directly: sentence length (`sentlen_z`) is retained as an additional nuisance covariate alongside mean word length and frequency, `roi` is a 12-level crossed random effect (the language-atlas parcellation used across all subjects), and surprisal is again computed over the whole sentence rather than word-by-word.

```
baseline: y ~ sentlen_z + mean_wlen_z + mean_freq_z +
                (1 + sentlen_z + mean_wlen_z + mean_freq_z
| subject) + (1 | roi) + (1 | sentence_id)
full: y ~ sentlen_z + mean_wlen_z + mean_freq_z +
surprisal_z +
           (1 + sentlen_z + mean_wlen_z + mean_freq_z +
surprisal_z | subject) + (1 | roi) +
           (1 + surprisal_z | sentence_id)
```

E.5 Intracranial recordings

For Podcast and Fedorenko2016, electrodes are patient-specific, so `electrode` is nested within `subject` rather than treated as a crossed effect. For Podcast, the model is fit separately at each lag of the response window; no spillover term is included.

```
baseline: y ~ wlen_z + freq_z + (1 | subject) + (1 |
subject:electrode)
full: y ~ wlen_z + freq_z + surprisal_z + (1 | subject)
+ (1 + surprisal_z | subject:electrode)
```

For Fedorenko2016, spillover is included, along with a richer by-subject random-slope structure:

```
baseline: y ~ wlen_z + freq_z + (1 + wlen_z + freq_z |
subject) +
                (1 | subject:electrode)
full: y ~ wlen_z + freq_z + surprisal_z +
prev_surprisal_z +
          (1 + wlen_z + freq_z + surprisal_z +
prev_surprisal_z | subject) +
          (1 + surprisal_z + prev_surprisal_z |
subject:electrode)
```

F Dependency Locality Theory Difficulty Stratification

To ask whether a model's advantage is uniform across words or concentrated on the words independently known to be difficult for incremental parsing, we stratify every word by its Dependency Locality Theory (DLT) integration cost (Gibson, 1998; 2000). DLT cost quantifies the working-memory demand of attaching a word to its syntactic head as a function of how much intervening discourse material – new discourse referents, chiefly nouns and verbs – has to be held active since the head was last processed: a word whose head is many discourse referents back costs more to integrate than one whose head was just seen. We compute this cost per word from a dependency parse of each dataset's stimuli, following Gibson's original definition directly rather than an approximation, and split words into an *easy* set (cost $= 0$: heads immediately available) and a *hard* set (cost ≥ 2, or a stricter cost $\geq$ 3–5 for the threshold sweep below), reusing the same BrainScore/BehavScore pipeline (Appendix D) but restricted to the corresponding word subset.

Figure 2B reports the hard set at cost ≥ 2. We chose this as the headline threshold after inspecting the full grid from cost ≥ 1 through ≥ 5: corpus-wide hard-word counts fall off sharply as the threshold rises (several behavioral corpora drop to single-digit word counts by cost $\geq$ 4–5, e.g. MECO, UCL Reading, and Provo), so a threshold has to be high enough to isolate genuinely hard integration cases but low enough that every dataset still contributes a usable sample. All datasets pool their hard-word items before averaging per model, so every hard-word comparison necessarily has less statistical power than the all-words comparison it is contrasted against, which is most visible in BehavScore's instability past cost ≥ 4.

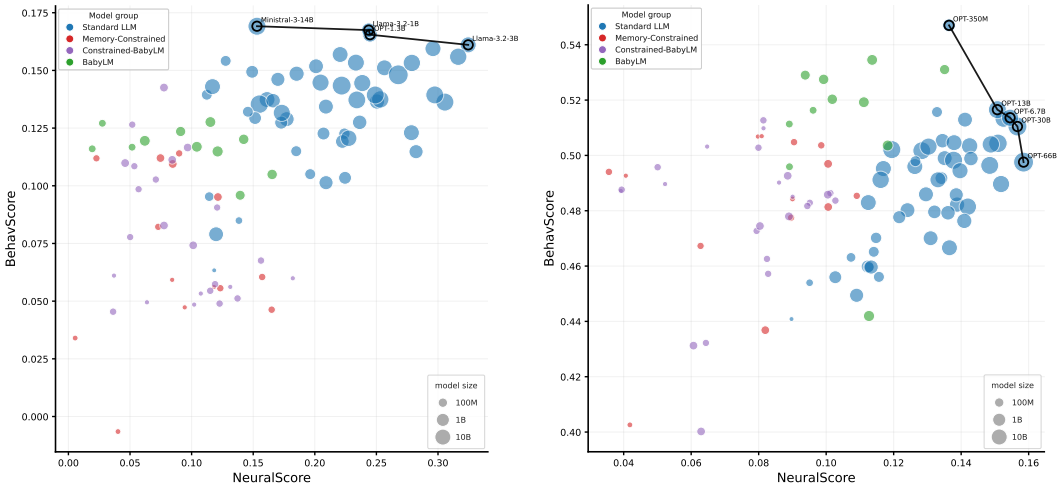
G Best-Layer Upper Bound

As noted in Section 3.3, we take each model's last layer as its primary summary score to avoid rewarding deeper models for having more opportunities to post a high score on held-out data by chance. Table 6 reports the alternative: each model's own best-scoring layer, averaged within group, for both tracks and both scores. The ordering between groups is unchanged from the last-layer results in Sections 4.1 and 4.3, and the gap between Standard LLMs and the two memory-constrained groups widens rather than narrows – consistent with our last-layer choice being the conservative comparison, not one that happens to favor our central claim.

Table 6: Mean best-layer NeuralScore and BehavScore by model group, both feature tracks.

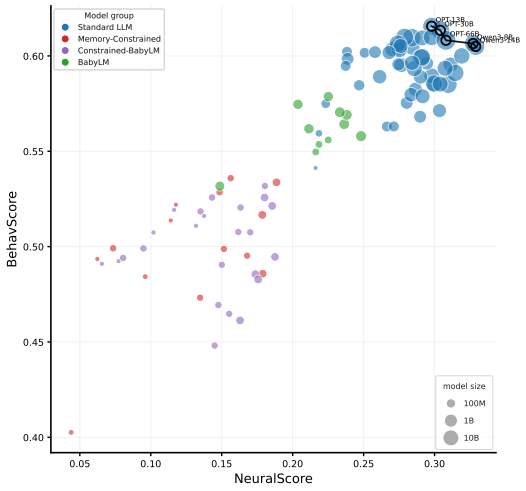
	Surprisal		Embedding	
Model group	NeuralScore	BehavScore	NeuralScore	BehavScore
Standard LLM	0.209	0.134	0.281	0.593
Memory-Constrained	0.090	0.070	0.129	0.499
Constrained-BabyLM	0.092	0.079	0.143	0.499
BabyLM	0.094	0.117	0.219	0.561

The Pareto front is similarly unchanged in composition (Figure 5). At best layer, the surprisal-track front is Llama-3.2-1B, Llama-3.2-3B, OPT-1.3B, and Ministral-3-14B; the embedding-track front is OPT-13B, OPT-30B, OPT-66B, Qwen3-8B, and Qwen3-14B; and the last-layer embedding front is OPT-350M, OPT-13B, OPT-6.7B, OPT-30B, and OPT-66B. Every model on every one of these fronts – and on the last-layer surprisal front of Figure 2A – is a Standard LLM.



(A) Surprisal, best layer.

(B) Embedding, last layer.



(C) Embedding, best layer.

Figure 5: Neural-behavioral Pareto fronts for the three (feature track, layer selection) combinations not shown in the main text (Figure 2A gives surprisal / last layer). Surprisal axes are DLL-ceiling-normalized; embedding axes are correlation-ceiling-normalized. Point size is model parameter count; color is model group; the black-outlined path traces the Pareto-optimal frontier. In all three, as in the main text, the frontier is occupied exclusively by Standard LLMs.

A caveat specific to the embedding track. Best-layer selection confounds two things for BrainScore that it does not confound for BehavScore. First, the layer-count bias already noted in Section 3.3: a deeper model gets more attempts at a high held-out score by chance. Second, and specific to embeddings, hidden-state width scales with model size alongside depth, so a larger model's ridge regression is also higher-dimensional – more regressors mechanically raise a ridge model's achievable in-sample fit, and while our nested cross-validation controls the regularization strength, it does not remove the advantage of having more (correlated) predictors to work with in the first place. Surprisal has neither confound: it is always a single scalar per layer regardless of a model's width, so the only thing best-layer selection can do for BehavScore is pick a better-fitting depth, not a wider one. This is also why we treat surprisal, not embeddings, as our primary comparison across model groups (Section 3.3) – the embedding track's NeuralScore gap between Standard LLMs and the constrained groups should be interpreted with this capacity confound in mind, rather than as evidence of representational alignment alone.

BabyLM's last-layer embedding dissociation. One pattern is specific to the last-layer embedding group means reported in Section 4.3 and is worth flagging on its own. At the last layer, BabyLM's mean embedding BehavScore (0.512) is higher than Standard LLM's (0.488; Welch's $t(13.0) = 2.87$, $p = 0.013$), while its mean embedding NeuralScore (0.105) is far lower than Standard LLM's (0.130; $t(16.1) = -5.19$, $p < 0.001$). In other words, a small model trained on a human-scale text budget can match or exceed a web-scale model at predicting reading times and fixations from its final-layer embedding, while remaining a substantially worse predictor of neural signal from that same embedding. This dissociation does not survive to the best layer (Table 6: Standard LLM leads BabyLM on both scores once every layer is available), so it is a property of what a small, data-constrained model's *last* representation looks like specifically, not of its representations in general – consistent with the layerwise finding (Section 4.4) that BabyLM tends to peak before its final layer.